Cole Scheller

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Education

Michigan State University — Sept. 2022 - May 2026
Bachelor of Science, Computer Engineering / Minor in Mathematics
Dean's list 2022-2025; '24-'25 College of Engineering Undergraduate Service Award
GPA: 3.5

Experience

CANVAS — Michigan State University — May 2023 - May 2026
Undergraduate Research Assistant

- Led a group labeling a subset of collected dataset samples to demonstrate the necessity of a multi-season, multi-modal dataset, contributing to the acceptance of the MSU-4S dataset at CVPR
- Modified the existing open-source point cloud annotation tool LabelCloud to meet the labeling team's needs by improving the pointcloud rotation algorithm, adding visual aids for interpreting the 3D scenes, and implementing additional keyboard shortcuts and bounding-box interaction methods, decreasing the labelling time per scene by approximately half
- Designed and executed an experiment to evaluate the use of existing dedicated short-range communications (DSRC) equipment for over-the-air sensor data sharing, supporting further development of cooperative perception for rail application

MCity — University of Michigan — May 2024 - Aug. 2024 & May 2025 - Aug. 2025
Engineering Intern

- Improved responsiveness of the facility's user-facing garage door control interface by replacing the existing polling-based design with event-based updates via MQTT's publish/subscribe model, deploying containers hosted through AWS's elastic container service (ECS) to run the MQTT broker and service, reducing garage door state update latency from around thirty seconds to at most two seconds
- Reduced the boot time of the facility's quasi-embedded real-time kinematic (RTK) units by nearly 25% by removing unnecessary steps from the existing U-Boot environment, as well as using systemd to identify and mitigate the most consequential startup jobs
- Replaced RTK units' manual update process with automated over-the-air updates by converting the units' services into Docker containers which are automatically built and uploaded to AWS's Elastic Container Registry (ECR) using AWS CodeBuild and CodePipeline, then configuring the units to retrieve the latest container images from ECR at startup
- Designed and implemented pytest-based testing infrastructure for internal API's legacy codebase, prioritizing flexibility to promote efficient as-needed test additions, allowing for a sustainably gradual adoption of testing within the project
- Integrated automated documentation into a legacy project using AWS CodeBuild and CodePipeline to independently generate Sphinx-based documentation upon repository updates without impacting existing source control and CI/CD workflows

Autonomous Vehicles Club — Michigan State University — Aug. 2022 - Apr. 2026
Transition Adviser (2026), President (2023-2025), Perception Team Lead (2023-2026)

- Facilitated leadership transition to incoming executive team while mentoring successors, improving year-end membership retention
- Worked with the leadership team to restore the club's pre-pandemic standards, coordinating student onboarding materials, social media outreach, and university administrative requirements.
- Developed RGB Camera-to-LiDAR point cloud projection in C++ using PCL and integrated with Robot Operating System (ROS), using TF2 to transform between LiDAR, camera, and world reference frames

ApprenTek — Dexter, MI — Jan. 2021 - Sept. 2021
IT Intern

FIRST Robotics Competition — Dexter, MI — Sept. 2018 - Apr. 2022
Team Captain (2020-2022), Computer Vision Team Lead (2018-2022)

Selected Projects

IndyTUI
A terminal-based telemetry display, designed to follow INDYCAR races live, with a configurable delay, or on a later date via recorded telemetry playback.

- Designed and implemented a thread-safe circular telemetry queue sized according to the telemetry refresh rate and desired delay, maintaining a bounded memory footprint for a given configuration
- Leveraged the telemetry queue's circular design and thread safety to update its head with newly received telemetry while only advancing it when the accrued delay is shorter than configured, prioritizing the delay's guarantee over telemetry continuity
- Deconstructed the telemetry API by monitoring a live local session with mitmproxy, identifying the endpoints and payloads required to initiate and maintain a session

AgentLang
A tree-walk interpreter for a simple scripting language used to define simulation-agent behavior across different simulation contexts. Developed for a capstone project.

- Implemented a two-stage parsing process that handles most parsing steps in the first stage and defers linking function-calls to their corresponding function-body AST nodes until finalization, enabling language features like recursion
- Designed and implemented the underyling simulation's agent action API using C++'s std::variant, enabling a straightforward visitor-based approach to action execution while decoupling action representation from the simulation context and agent type

LabelCloud
A fork of a PyQt-based point cloud annotation tool, extended to support camera extrinsic calibration.

- Significantly narrowed the scope of strategies within the existing strategy-based design, allowing a greater number of UI/UX elements to function independently of usage-mode
- Reorganized the interface and used QT DynamicProperties to control event handling and mode-dependent behavior, improving extensibility by decreasing the amount of mode-dependent initialization and event logic
- Added an interactive image display to the main interface which could support zooming, panning, and manual pixel selection, enabling manual correspondence between point cloud points and image points

Skills

Languages: C++, Python, Bash, GNU Emacs Lisp, LaTeX
Frameworks/Libraries: NumPy, OpenCV, Flask, ROS 1, ROS 2, TF2, MQTT, Catch2, notcurses
Cloud & DevOps: AWS ECS, CodePipeline, CodeBuild, Docker, Git, Linux